Jatin Bharati

j18bharati@gmail.com | 4076903578 | San Francisco, CA | linkedin.com/in/jatin-bharati | github.com/J18Bharati

Professional Summary

Software Engineer with expertise in generative AI, large language models (LLMs), model alignment, reinforcement learning, and computer vision. Strong foundation in mathematics and statistics, eager to leverage advanced my skills in collaborative environments to deliver cutting-edge solutions that enhance customer experiences and drive business value.

Open to relocation nationwide.

Technical Skills

- **Programming Languages:** Python, C, C++, Java, SQL, R, MATLAB, JavaScript, Rust
- **Frameworks & Libraries:** NumPy, Pandas, Scikit-learn, PyTorch, OpenCV, Hugging Face Transformers, FastAPI
- **AI/ML Specializations:** Reinforcement Learning, Computer Vision, Deep Neural Networks, Statistical Modeling, LLMs/SLMs, Fine-Tuning, Model Alignment
- **Tools & Technologies:** Git, AWS Bedrock, Docker, Databricks, Debugging Tools, CI/CD Pipelines

Certifications

Machine Learning Engineering for Production (MLOps) Specialization

DeepLearning.AI June 2024

- Comprehensive coverage of production ML systems, deployment strategies, and MLOps practices, including model monitoring and responsible AI implementation
- **Skills:** Model deployment, monitoring, CI/CD for ML, data and model management, quality assurance
- **Verification:** coursera.org/verify/TBNZRX3AUGTF

Education

University of Central Florida

Master of Science in Computer Science August 2023 - May 2025

- **Relevant Coursework:** Machine Learning, Computer Vision, Mathematical Statistics, Mathematical Foundations for Massive Data, Advanced Database Systems, Advanced Data Structures, Machine Learning for Biomedical Data
- Focused on **AI Model Alignment** and **Machine Learning Applications**, including evaluation of AI technologies like LLMs and SLMs.
- Served as **Coordinator for AI@UCF**, leading research discussions on fixing hallucinations in vision-language models (VLLMs), incorporating responsible AI practices.

Birla Institute of Technology and Science, Pilani

Integrated Master of Science in Mathematics August 2018 - August 2023

- **Relevant Coursework:** Applied Statistical Methods, Optimization, Discrete Mathematics, Numerical Analysis, Graphs and Networks, Applied Stochastic Process, Number Theory, Cryptography
- Conducted thesis research on **elliptic curve cryptography** (ECC), investigating algebraic structures, finite fields, and bilinear pairings to develop enhanced identity-based encryption schemes and **post-quantum cryptographic** solutions.
- Collaborated with Center for Entrepreneurial Leadership (CEL) at BITS Goa as Core Member and Crew Member, organizing events and summits while facilitating connections with startup founders, entrepreneurs, and industry executives to foster engagement and gain strategic insights

Experience

Software Engineer – AI & DevOps

Zof AI
Internship July 2025 - August 2025

- Developed and maintained backend and full-stack features incorporating LLM/SLM agents and self-healing algorithms for real-time deployment, ensuring scalability and low-latency inference.
- Architected scalable systems to support model pipelines, secure data handling, and performance expectations while applying best practices for security, privacy, and accessibility.
- Collaborated with stakeholders and AI/ML teams to define feature requirements, integrate feedback loops, and enhance designs based on customer metrics.
- Optimized system reliability, speed, and deployment workflows using Docker, cloud platforms, and CI/CD pipelines, proactively addressing issues with debugging tools and logs.

Software Engineer

MyTask Software
Internship May 2022 - July 2022

- Developed and implemented key software solutions that streamlined processes and improved operational efficiency.
- Designed an AI-based document classification system, reducing processing time by 40%.
- Improved financial compliance prediction accuracy by 20% using logistic regression and support vector machines.
- Created AI-powered Python tools and dashboards for client management, enhancing document classification and operational efficiency.
- Conducted API integration testing and code reviews, gaining insights into production software development practices.

Summer Researcher

APHRDI
Internship May 2021 - July 2021

- Conducted systematic review of 15+ research papers on AI technologies, translating complex concepts into actionable insights for technical teams.
- Developed concise research reports showcasing ability to analyze and communicate technical information effectively, with a focus on AI/ML advancements.
- Presented findings to stakeholders, demonstrating strong communication and analytical skills while incorporating feedback for continuous improvement.

Projects

Hybrid Language Model Enrichment

- Fine-tuned GPT-2 (SLM) using reinforcement learning to enhance language model performance and alignment.
- Designed innovative solution to enrich frozen LLM outputs with customized GPT-2 token integration, evaluating architectures like orchestration patterns.
- Implemented guided output strategies using fine-tuned tokens to align generated content with specific sentiment targets, improving accuracy scores of GPT-4o-mini by 40% while applying responsible AI practices.

Real-Time Object Tracking System

- Designed and implemented advanced computer vision solution using Convolutional Neural Networks (CNNs) for object detection and tracking in video frames.
- Developed sophisticated tracking algorithm with IoU (Intersection over Union) method to predict object locations, achieving real-time performance.
- Leveraged Greedy Matching to reduce tracking latency by approximately 15 times, and implemented motion prediction to account for skipped frames, increasing recall by 20%.
- Achieved near-perfect precision of approximately 97% for object localization and tracking, ensuring scalability for production environments.

Polygenic Trait Prediction Using Deep Learning

- Processed and filtered genomic data from openSNP, creating refined dataset of 455 samples for machine learning analysis.
- Leveraged PredictDB and GTEx v8 models for gene expression level prediction in the musculoskeletal system, identifying 5,486 unique genetic variants.
- Implemented Deep Neural Network (DNN) architecture featuring skip connections to enhance predictive capability for adult height prediction.
- Achieved cutting-edge accuracy of 30% within 2% error margin despite limited data, demonstrating potential of gene expression-based prediction with fewer input features compared to traditional genome-wide association studies.